디지털 콘텐츠 개론

PC와 모바일을 위한 콘텐츠

Contents Processing for PC and Mobile

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다양한 단말기의 등장

- PC
  - Full HD, HD, UHD, QWHD
- 모바일 단말기
  - iPhone, 갤럭시
  - iPad, iPad Mini
  - 갤럭시 탭
  - 듀얼 모니터 모바일 단말기
- 사용자가 어떤 기기로 접속하든 최적화하자.

다양한 단말기의 등장

- 과거
  - 각 사이즈별로 계산해 각기 다른 HTML 작업
  - 유지 보수의 심각한 문제
- 현재
  - 하나의 코드로 단말기 사이즈별 동작 표시
  - 부트스트랩5
    - https://www.w3schools.com/bootstrap5/

부트스트랩5(Bootstrap 5)

- 스타일을 단순화
  - 색상 표현
    - primary, danger, success, warning, secondary, info, …
    - 자주 사용하는 범용 색상
    - 배경색 : bg-primary, bg-danger, …
    - 글씨색 : text-primary, text-white, text-danger, …
  - 단말기 사이즈
    - xs, sm, md, lg, xlg, xxlg, ..
  - 화면을 12 칸으로 구분

Bootstrap5 기본 코드

- Bootstrap5 기본 코드

당신

HTML5에 bootstrap5를 사용하기 위한 기본 코드를 만들어줘.

ChatGPT

아래는 HTML5에서 Bootstrap 5를 사용하기 위한 기본 코드를 만든 예제입니다. 이 예제는 Bootstrap의 CDN을 통해 CSS 및 JavaScript 라이브러리를 불러오고, Bootstrap의 기본 스타일 및 기능을 사용할 수 있도록 합니다.

html Copy code

```html
<!DOCTYPE html>
<html lang="ko">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Bootstrap 5 Example</title>
<!-- Bootstrap CSS -->
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0-alpha1/dist/css/bootstrap.mi
</head>
<body>

<!-- 내용 -->
<div class="container">
    <h1>Hello, Bootstrap 5!</h1>
    <p>This is a basic example of using Bootstrap 5 in HTML5.</p>
</div>

<!-- Bootstrap JavaScript 및 의존성 라이브러리 -->
<script src="https://cdn.jsdelivr.net/npm/@popperjs/core@2.11.6/dist/umd/popper.min.j
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0-alpha1/dist/js/bootstrap.mi
</body>
</html>
```

Grid 시스템의 이해(가로

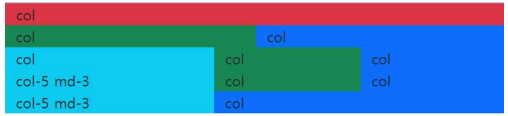
- Grid System
- 12칸

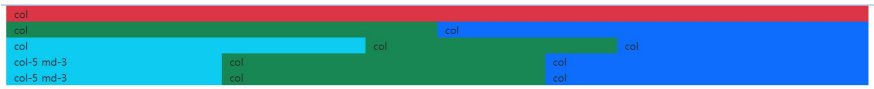
```
10  <body>
11
12  <div class="container">
13      <div class="row">
14          <div class="col bg-danger">col</div>
15      </div>
16
17      <div class="row">
18          <div class="col bg-success">col</div>
19          <div class="col bg-primary">col</div>
20      </div>
21
22      <div class="row">
23          <div class="col-5 bg-info">col</div>
24          <div class="col bg-success">col</div>
25          <div class="col bg-primary">col</div>
26      </div>
27
28      <div class="row">
29          <div class="col-5 col-md-3 bg-info">col-5 md-3</div>
30          <div class="col bg-success">col</div>
31          <div class="col bg-primary">col</div>
32      </div>
33
34      <div class="row">
35          <div class="col-5 col-md-3 bg-info">col-5 md-3</div>
36          <div class="col d-none d-md-block bg-success">col</div>
37          <div class="col bg-primary">col</div>
38      </div>
39  </div>
40
41  </body>
```

container

col(컬럼)

row(라인)





style 정의 (style.css)

```
/* style.css */
html, body { max-width:100%; overflow-x:hidden;}
table { border-collapse: collapse; }

a:link { text-decoration:none;  color:#333333; }
a:hover { text-decoration:none;  color:#0000FF; font-weight:bold; }
a:visited { text-decoration:none;  color:#333333; }

.ellipsis {
    white-space: nowrap;
    overflow: hidden;
    text-overflow: ellipsis;
    max-width:0;
}

.icon{ vertical-align:text-bottom; }

.row { line-height:200%; min-width:100%; min-height:30px; height:auto; padding-top:5px;
border:0px solid #FFFFFF; }
.colLine { line-height:200%; min-height:30px; border-bottom:1px dotted #333333; padding-top:4px;
padding-bottom:4px;}
```

```css
/* style.css */

html, body { max-width:100%; overflow-x:hidden;}
table { border-collapse: collapse; }
a:link { text-decoration:none; color: #333333; }
a:hover { text-decoration:none; color: #0000FF; font-weight:bold; }
a:visited { text-decoration:none; color: #333333; }
.ellipsis {
    white-space: nowrap;
    overflow: hidden;
    text-overflow: ellipsis;
    max-width:0;
}
.icon{ vertical-align:text-bottom; }
.row {
    line-height:200%;
    min-width:100%;
    min-height:30px;
    height:auto;
    padding-top:5px;
    border:0px solid #FFFFFF;
}
.colLine {
    line-height:200%;
    min-height:30px;
    border-bottom:1px dotted #333333;
    padding-top:4px; padding-bottom:4px;
}
```

```html
<!DOCTYPE html>
<html lang="ko">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Bootstrap 5 그리드 시스템 예제</title>
<!-- Bootstrap CSS -->
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0-alpha1/dist/css/bootstrap.min.css" rel="stylesheet">
<link rel="stylesheet" href="style.css">
</head>
<body>

<div class="container">
    <div class="row">
        <div class="col colLine">col</div>
    </div>
    <div class="row">
        <div class="col colLine text-primary">Primary</div>
        <div class="col colLine text-danger">Danger</div>
        <div class="col colLine text-success">Success</div>
    </div>
    <div class="row">
        <div class="col colLine text-primary text-start">Primary Start</div>
        <div class="col colLine text-danger text-center">Danger Center</div>
        <div class="col colLine text-success text-end">Success End</div>
    </div>

</body>
</html>
```

스타일을 파일로 연결

External CSS

왼쪽 정렬 : text-start (기본)

중앙 정렬 : text-center

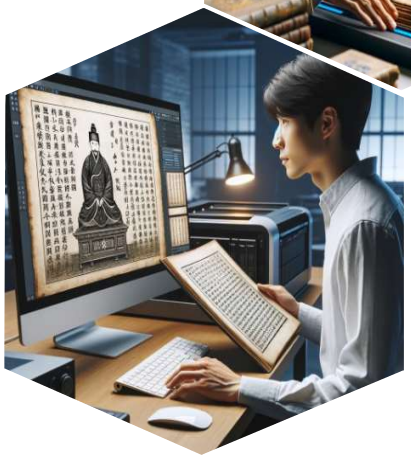
우측 정렬 : text-end

col

Primary	Danger	Success
Primary Start	Danger Center	Success End

텍스트 / 이미지 콘텐츠

- 텍스트 출력 공간 부족
  - 스마트폰의 세로 상태
  - 긴 글은 말줄임…
- 이미지 크기
  - 너무 크면 확대했을 때 깨짐 현상
  - 스마트 폰에서 이미지를 화면에 맞게



1

Bootstrap 핵심 기능

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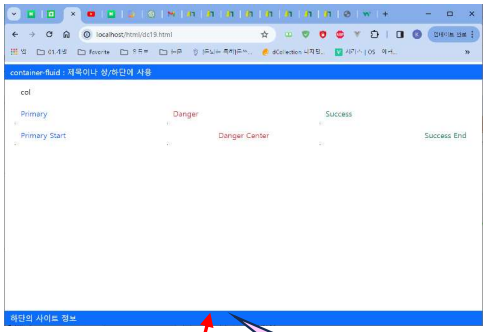
컨테이너

-
  - 가로 : 화면 일부, 좌우 여백
-
  - 가로 : 화면 전체

```html
<div class="container-fluid">
    <div class="row">
        <div class="col bg-primary text-white">
            container-fluid : 제목이나 상/하단에 사용
        </div>
    </div>
</div>
<div class="container">
    <div class="row">
        <div class="col colLine">col</div>
    </div>
    <div class="row">
        <div class="col colLine text-primary">Primary</div>
        <div class="col colLine text-danger">Danger</div>
        <div class="col colLine text-success">Success</div>
    </div>
    <div class="row">
        <div class="col colLine text-primary text-start">Primary Start</div>
        <div class="col colLine text-danger text-center">Danger Center</div>
        <div class="col colLine text-success text-end">Success End</div>
    </div>
</div>
<div class="container-fluid" id="footer">
    <div class="row">
        <div class="col bg-primary text-white">
            하단의 사이트 정보
        </div>
    </div>
</div>
```



맨 아래에!!!

```style
#footer
{
    position:fixed; bottom:0; width:100%;
}
```

테이블 (w3schools.com/bootstra

- 테이블
- (w3schools.com/bootstrap5 참고)

Hover Rows

The .table-hover class enables a hover state (grey background on mouse over) on table rows:

Firstname	Lastname	Email
John	Doe	john@example.com
Mary	Moe	mary@example.com
July	Dooley	july@example.com

mouse over

Hover Rows

The .table-hover class enables a hover state (grey background on mouse over) on table rows:

Firstname	Lastname	Email
John	Doe	john@example.com
Mary	Moe	mary@example.com
July	Dooley	july@example.com

see @next page

```
11  <body>
12      <div class="container mt-3">
15          <table class="table table-hover">
16            <thead>
18              <th>Firstname</th>
19              <th>Lastname</th>
20              <th>Email</th>
21            </tr>
22            </thead>
23            <tbody>
24              <tr>
25              <td>John</td>
26              <td>Doe</td>
27              <td>john@example.com</td>
28              </tr>
29              <tr>
30              <td>Mary</td>
31              <td>Moe</td>
32              <td>mary@example.com</td>
33              </tr>
34              <tr>
35              <td>July</td>
36              <td>Dooley</td>
37              <td>july@example.com</td>
38              </tr>
39            </tbody>
40          </table>
41        </div>
42  </body>
```

BS5 Utilities

BS5 Tables
BS5 Images
BS5 Jumbotron
BS5 Alerts
BS5 Buttons
BS5 Button Groups
BS5 Badges
BS5 Progress Bars
BS5 Spinners
BS5 Pagination
BS5 List Groups
BS5 Cards
BS5 Dropdowns
BS5 Collapse
BS5 Navs
BS5 Navbar
BS5 Carousel
BS5 Modal
BS5 Tooltip
BS5 Popover
BS5 Toast
BS5 Scrollspy
BS5 Offcanvas
BS5 Utilities

Where *property* is one of:

- `m` - sets `margin`
- `p` - sets `padding`

Where *sides* is one of:

- `t` - sets `margin-top` or `padding-top`
- `b` - sets `margin-bottom` or `padding-bottom`
- `s` - sets `margin-left` or `padding-left`
- `e` - sets `margin-right` or `padding-right`
- `x` - sets both `padding-left` and `padding-right` or `margin-left` and `margin-right`
- `y` - sets both `padding-top` and `padding-bottom` or `margin-top` and `margin-bottom`
- blank - sets a `margin` or `padding` on all 4 sides of the element

Where *size* is one of:

- `0` - sets `margin` or `padding` to `0`
- `1` - sets `margin` or `padding` to `.25rem`
- `2` - sets `margin` or `padding` to `.5rem`
- `3` - sets `margin` or `padding` to `1rem`
- `4` - sets `margin` or `padding` to `1.5rem`
- `5` - sets `margin` or `padding` to `3rem`
- `auto` - sets `margin` to auto

List

List

- unordered list 1
- unordered list 2
- unordered list 3

1. ordered list 1
2. ordered list 2
3. ordered list 3

```html
<body>
  <div class="container">
    <h2 class="text-primary">List</h2>
    <div class="row">
      <div class="col colLine">
        <ul>
          <li>unordered list 1</li>
          <li>unordered list 2</li>
          <li>unordered list 3</li>
        </ul>
      </div>
    </div>
    <div class="row">
      <div class="col colLine">
        <ol>
          <li>ordered list 1</li>
          <li>ordered list 2</li>
          <li>ordered list 3</li>
        </ol>
      </div>
    </div>
  </div>
</body>
```

BS5 List

```html
<body>
  <div class="container">
    <h2 class="text-primary">Pagination</h2>

    <div class="row">
      <div class="col colLine">
        <ul class="pagination">
          <li class="page-item"><a class="page-link" href="#">Previous</a></li>
          <li class="page-item"><a class="page-link" href="#">1</a></li>
          <li class="page-item active"><a class="page-link" href="#">2</a></li>
          <li class="page-item"><a class="page-link" href="#">3</a></li>
          <li class="page-item"><a class="page-link" href="#">Next</a></li>
        </ul>
      </div>
    </div>
  </div>
</body>
```

Pagination

Previous 1 2 3 Next

아이콘 : Material-icons

당신

material-icons를 이용하기 위한 HTML5의 헤더를 포함한 예제를 보여줘.

ChatGPT

Material Icons를 사용하기 위해서는 Google의 Material Icons 라이브러리를 불러와야 합니다. 아래는 Material Icons를 사용하기 위한 HTML5의 헤더를 포함한 예제 코드입니다.

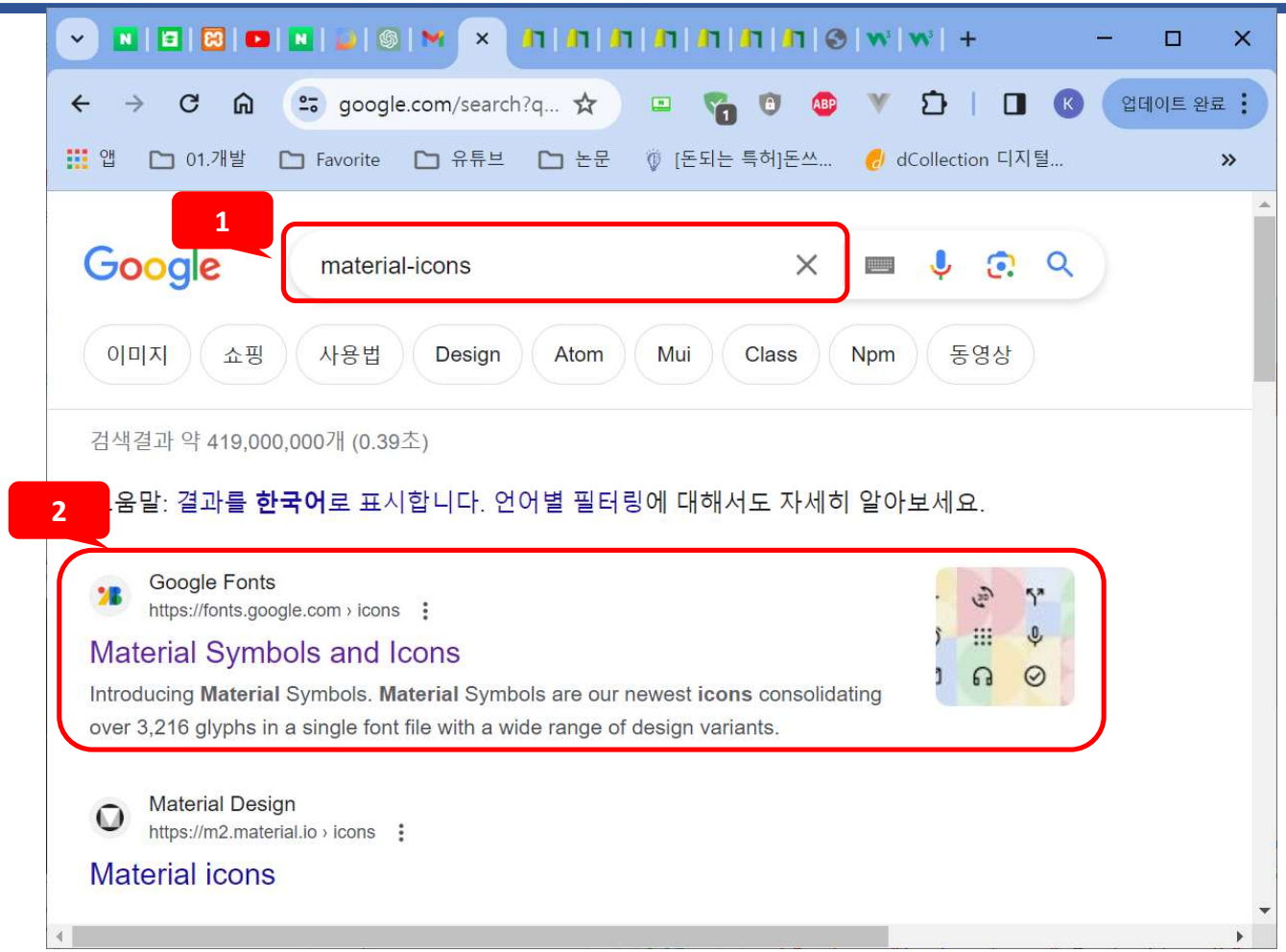
html Copy code

```html
<!DOCTYPE html>
<html lang="ko">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Material Icons 예제</title>
<!-- Google Material Icons -->
<link href="https://fonts.googleapis.com/icon?family=Material+Icons" rel="stylesheet"
</head>
<body>

<!-- Material Icons 사용 예제 -->
<p>여기에 Material Icons를 사용할 수 있습니다: <span class="material-icons">favorite</s

</body>
</html>
```

Google : Material-icons

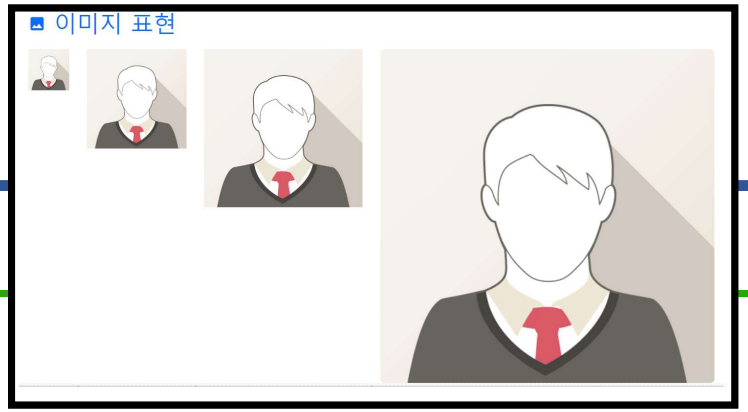


```html
<link href="https://fonts.googleapis.com/icon?family=Material+Icons" rel="stylesheet">
<link rel="stylesheet" href="style.css">
</head>
<body>
  <div class="container">
    <h2 class="text-primary">
        <span class="material-icons">settings</span>Pagination
    </h2>

    <div class="row">
        <div class="col colLine">
            <ul class="pagination">
            <li class="page-item">
                <a class="page-link" href="#">
                    <span class="material-icons">chevron_left</span>
                </a>
            </li>
            <li class="page-item"><a class="page-link" href="#">1</a></li>
            <li class="page-item active"><a class="page-link" href="#">2</a></li>
            <li class="page-item"><a class="page-link" href="#">3</a></li>
            <li class="page-item">
                <a class="page-link" href="#">
                    <span class="material-icons">chevron_right</span>
                </a>
            </li>
            </ul>
        </div>
    </div>
  </div>
</body>
```

마음에 드는 아이콘 선택

BS5 : 이미지




```html
<body>
  <div class="container">
    <h2 class="text-primary">
        <span class="material-icons">image</span> 이미지 표현
    </h2>

    <div class="row">
        <div class="col-1 colLine">
            <img src="data/images/img_avatar1.png" class="img-fluid">
        </div>
        <div class="col-2 colLine">
            <img src="data/images/img_avatar1.png" class="img-fluid">
        </div>
        <div class="col-3 colLine">
            <img src="data/images/img_avatar1.png" class="img-fluid">
        </div>
        <div class="col colLine">
            <img src="data/images/img_avatar1.png" class="img-fluid rounded">
        </div>
    </div>
  </div>
</body>
```

BS5 > Card에서 다운로드

부트스트랩 정리

- PC와 모바일을 한번에 만들 수 있는 라이브러리
  - 격자(Grid) 시스템 : 12칸
  - 자주 사용하는 색상의 텍스트화
    - primary, danger, success, info, warning
    - 배경색 : bg-primary
    - 글씨색 : text-danger
  - 아이콘화 : 모바일에 적합
    - 직관적
    - material-icons